**Department of Mathematics, IIT Madras**

**MA2101 Linear Algebra via Matrices**

**Assignment 4**

1. Compute $\langle (2,1,3),(1,3,2)\rangle$ and $\langle (1,2i,3),(1+i,i,-3i)\rangle$.
2. Is $\{[1,2,3,-1]^t,\ [2,-1,0,0]^t,\ [0,0,1,3]^t\}$ an orthogonal set in $\mathbb{F}^{4\times 1}$?
3. Show that if $x, y \in \mathbb{F}^n$, then $x \perp y$ implies $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.
4. Show that if $x, y \in \mathbb{R}^n$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ implies $x \perp y$.
5. In $\mathbb{C}$, the inner product is given by $\langle x, y\rangle = \bar{y}x$. Let $x = 1$ and $y = i$ be two vectors in $\mathbb{C}$. Show that $\|x\|^2 + \|y\|^2 = \|x+y\|^2$ but $\langle x, y\rangle \neq 0$.
6. Show that the *parallelogram law* holds in $\mathbb{F}^n$. That is, for all $x, y \in \mathbb{F}^n$, we have $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$.
7. If $x$ and $y$ are orthogonal vectors in $\mathbb{F}^n$, then show that $\|x\| \leq \|x+y\|$.
8. Construct an orthonormal set from $\{[1,2,0]^t,\ [2,-1,0]^t,\ [0,0,2]^t\}$.
9. Orthogonalize the vectors $(1,1,1)$, $(1,0,1)$, and $(0,1,2)$ using Gram-Schmidt process.
10. How do we use Gram-Schmidt to compute the rank of a matrix?
11. Show that the cross product $u \times v$ of two linearly independent vectors $u, v$ in $\mathbb{R}^{1\times 3}$ is orthogonal to both $u$ and $v$. How to obtain this third vector as $u \times v$ by Gram-Schmidt process?
12. Find $u \in \mathbb{R}^{1\times 3}$ so that $\{[1/\sqrt{3},\ 1/\sqrt{3},\ 1/\sqrt{3}],\ [1/\sqrt{2},\ 0,\ -1/\sqrt{2}],\ u\}$ is an orthonormal set. Form a matrix with the vectors as rows, in that order. Verify that the columns of the matrix are also orthonormal.
13. Let $\{u, v\}$ be an orthonormal basis for $\mathbb{R}^{2\times 1}$. Let $x \in \mathbb{R}^{2\times 1}$ be a unit vector. If $x^t u = \frac{1}{2}$, then what could be $x^t v$?
14. Let $A \in \mathbb{F}^{m\times n}$. Show the following:
    (a) For $m \leq n$, $AA^* = I$ iff the rows of $A$ are orthonormal.
    (b) For $m \geq n$, $A^*A = I$ iff the columns of $A$ are orthonormal.
15. Find a QR-factorization of each of the following matrices:

$$\text{(a)}\ \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \qquad \text{(b)}\ \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \qquad \text{(c)}\ \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

16. Let $\{u, v, w\}$ be an orthonormal basis for a subspace of $\mathbb{R}^{5\times 1}$ and let $x = au+bv+cw$. If $\|x\| = 5$, $\langle x, u\rangle = 4$ and $x \perp v$, then what are the possible values of $a$, $b$ and $c$?
17. Let $A$ be the $4 \times 2$ matrix whose rows are $[1/2,\ -1/2]$, $[1/2,\ 1/2]$, $[1/2,\ -1/2]$ and $[1/2,\ 1/2]$.
    (a) Determine the projection matrix $P$ that projects vectors onto $R(A)$.
    (b) Find an orthonormal basis for $N(A^t)$, and then determine the projection matrix $Q$ that projects vectors onto $N(A^t)$.
    (c) Is it true that $P^2 = P^t = P$, and $Q^2 = Q^t = Q$?
18. Let $y$ be the orthogonal projection of a vector $x \in \mathbb{F}^{n\times 1}$ onto a subspace $V$ of $\mathbb{F}^{n\times 1}$. Show that $\|y\|^2 = \langle x, y\rangle$.
19. Find the best approximation of $x \in U$ from $V$ where
    (a) $U = \mathbb{R}^3$, $x = (1,2,1)$, $V = \text{span}\{(3,1,2),(1,0,1)\}$
    (b) $U = \mathbb{R}^3$, $x = (1,2,1)$, $V = \{(\alpha,\beta,\gamma) \in \mathbb{R}^3 : \alpha+\beta+\gamma = 0\}$
    (c) $U = \mathbb{R}^4$, $x = (1,0,-1,1)$, $V = \text{span}\{(1,0,-1,1),(0,0,1,1)\}$
20. Find least squares solution(s) of the system $Ax = b$, where

$$\text{(a)}\ A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \\ 2 & -1 \end{bmatrix},\ b = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \qquad \text{(b)}\ A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix},\ b = \begin{bmatrix} 0 \\ 1 \\ -1 \\ -2 \end{bmatrix}$$